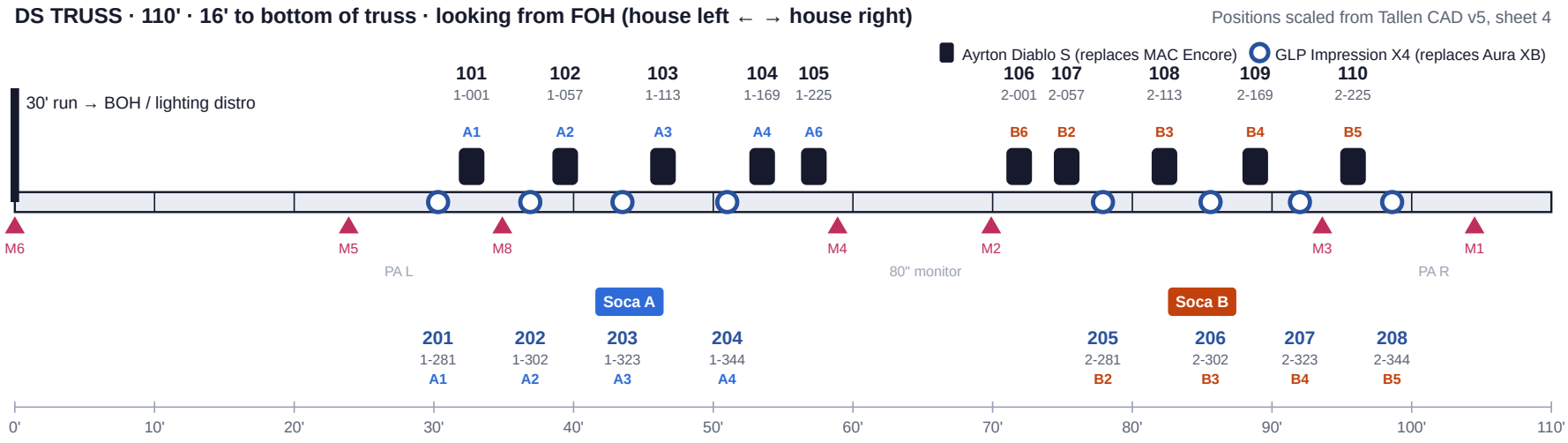
NIXON PEABODY · HYATT REGENCY CHICAGO · GRAND BALLROOM · OCT 14–17, 2026

Lighting Power & Circuit Plan

digerati

FIXTURES	TOTAL LOAD (MAX)	PER LEG · 208Y/120	SOCAPEX RUNS	SOCAPEX (WITH SPARES)
44	**14.4 kW**	**41 A max**	**4**	**700'**
10 Diablo S · 18 X4 · 16 COLORado	119.8 A @ 120 V	A 39.5 · B 39.5 · C 40.7	19 circuits used · 5 spare	4 × 100' · 5 × 50' · 2 × 25' · 525' in the plan

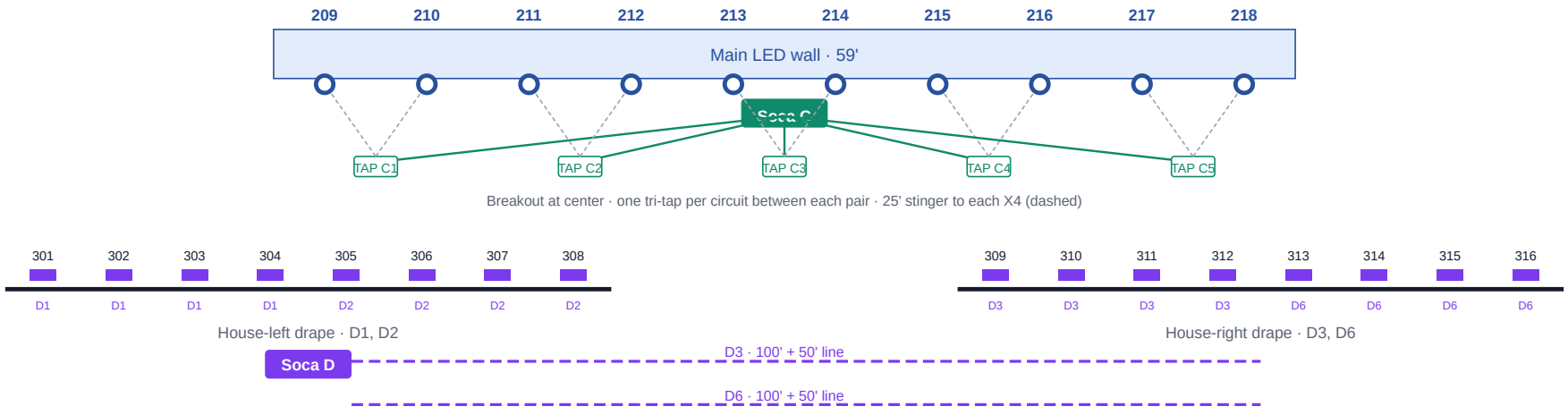
DOWNSTAGE TRUSS PLOT



Planning rules: 120 V circuits on 20 A breakers, loaded to no more than 16 A (80% continuous). Wattages are manufacturer maximums: Diablo S 550 W · Impression X4 350 W · COLORado Batten 72X 161 W. Soca circuits 1–6 assumed wired to phases A-B-C-A-B-C at the distro. Fixtures power-link (TRUE1 / powerCON) within a circuit. Truss positions are scaled from the CAD symbols and should be confirmed at hang.

LED WALL & FLOOR

LED WALL X4s (Soca C) · FLOOR COLORADOS (Soca D) · looking from FOH



SOCAPEX RUNS

RUN	FEEDS	ROUTE	DISTANCE	CABLE
Soca A	DS truss, house-left half (U1)	Along DS truss to HL end · 44' 30' truss run toward BOH · 30' Drop to floor (16' trim + dress) · 18' Floor to lighting distro (BOH) · 17'	109' 131' w/ 20%	**100' + 50'** = 150'
Soca B	DS truss, house-right half (U2)	Along DS truss to HL end · 85' 30' truss run toward BOH · 30' Drop to floor (16' trim + dress) · 18' Floor to lighting distro (BOH) · 17'	150' 180' w/ 20%	**100' + 50' + 50'** = 200'
Soca C	LED wall X4s (U3)	Distro to upstage center, behind LED wall · 85' Up to the wall (~15') · 15'	100' 120' w/ 20%	**100' + 25'** = 125'
Soca D	All 16 floor COLORados	Distro to HL drape midpoint · 40'	40' 48' w/ 20%	**50'** = 50'

Run lengths are estimates from the CAD (1/8" = 1'). They're measured from the lighting distro by the 200 A camlock backstage (house left), up the 30' truss run and along the DS truss, with a 16' drop at trim, then **20% overage** and rounded up to whole cable lengths. Soca B is the longest run at ~200': expect about 5% voltage drop at 7.5 A on 14 AWG. That's OK for these auto-ranging fixtures, but don't add load to those circuits.

CIRCUIT SCHEDULE

CIRCUIT	FIXTURES (UNIT #)	LOAD	PHASE	AMPS
Soca A · DS truss, house-left half (U1) · breakout: On DS truss at ~44' from the HL end · 4,150 W				
A1	201 (X4), 101 (Diablo)	900 W	A	7.5 A
A2	202 (X4), 102 (Diablo)	900 W	B	7.5 A
A3	203 (X4), 103 (Diablo)	900 W	C	7.5 A
A4	204 (X4), 104 (Diablo)	900 W	A	7.5 A
A6	105 (Diablo)	550 W	C	4.6 A
Spare: A5				
Soca B · DS truss, house-right half (U2) · breakout: On DS truss at ~85' from the HL end · 4,150 W				
B2	107 (Diablo), 205 (X4)	900 W	B	7.5 A
B3	108 (Diablo), 206 (X4)	900 W	C	7.5 A
B4	109 (Diablo), 207 (X4)	900 W	A	7.5 A
B5	110 (Diablo), 208 (X4)	900 W	B	7.5 A
B6	106 (Diablo)	550 W	C	4.6 A
Spare: B1				
Soca C · LED wall X4s (U3) · breakout: Behind LED wall, dead center. Circuits fan out left and right · 3,500 W				
C1	209 (X4), 210 (X4)	700 W	A	5.8 A
C2	211 (X4), 212 (X4)	700 W	B	5.8 A
C3	213 (X4), 214 (X4)	700 W	C	5.8 A
C4	215 (X4), 216 (X4)	700 W	A	5.8 A
C5	217 (X4), 218 (X4)	700 W	B	5.8 A
Spare: C6				
Soca D · All 16 floor COLORados · breakout: Midpoint of the house-left drape line · 2,576 W				
D1	301 (COLORado), 302 (COLORado), 303 (COLORado), 304 (COLORado)	644 W	A	5.4 A
D2	305 (COLORado), 306 (COLORado), 307 (COLORado), 308 (COLORado)	644 W	B	5.4 A
D3	309 (COLORado), 310 (COLORado), 311 (COLORado), 312 (COLORado)	644 W	C	5.4 A
D6	313 (COLORado), 314 (COLORado), 315 (COLORado), 316 (COLORado)	644 W	C	5.4 A
Spare: D4, D5				

CABLE PULL

ITEM	IN THE PLAN	SPARE	PULL
Socapex 19-pin, 100'	3	1	**4**
Socapex 19-pin, 50'	4	1	**5**
Socapex 19-pin, 25'	1	1	**2**
Total Socapex	525'	175'	**700'**
Soca → 6 × Edison breakout	4	1	**5**
12/3 Edison extension, 100' (Soca D cross-stage)	2	0	**2**
12/3 Edison extension, 50' (Soca D cross-stage)	2	0	**2**
Tri-tap (LED wall)	5	1	**6**
25' Edison stinger (LED wall)	10	2	**12**

Socapex and breakouts fit the PRG order (4 × 100', 6 × 50', 4 × 25', 6 breakouts) with spares. The **12/3 extensions, tri-taps and 25' stingers** aren't on any order yet. Pull them from Digerati stock or add them to PRG. X4s take powerCON, so the stingers land on the Edison → powerCON tails that come with the Rentex X4 kit.

PATCH

UNIT	FIXTURE	POSITION	ON TRUSS	DMX (U-ADDR)	MODE	CIRCUIT
101	Ayrton Diablo S	DS truss	32.7'	1-001	Diablo Ex · 56 ch	A1
102	Ayrton Diablo S	DS truss	39.4'	1-057	Diablo Ex · 56 ch	A2
103	Ayrton Diablo S	DS truss	46.4'	1-113	Diablo Ex · 56 ch	A3
104	Ayrton Diablo S	DS truss	53.5'	1-169	Diablo Ex · 56 ch	A4
105	Ayrton Diablo S	DS truss	57.2'	1-225	Diablo Ex · 56 ch	A6
106	Ayrton Diablo S	DS truss	71.9'	2-001	Diablo Ex · 56 ch	B6
107	Ayrton Diablo S	DS truss	75.3'	2-057	Diablo Ex · 56 ch	B2
108	Ayrton Diablo S	DS truss	82.3'	2-113	Diablo Ex · 56 ch	B3
109	Ayrton Diablo S	DS truss	88.8'	2-169	Diablo Ex · 56 ch	B4
110	Ayrton Diablo S	DS truss	95.8'	2-225	Diablo Ex · 56 ch	B5
201	GLP Impression X4	DS truss · room wash	30.3'	1-281	Extended · 21 ch	A1
202	GLP Impression X4	DS truss · room wash	36.9'	1-302	Extended · 21 ch	A2
203	GLP Impression X4	DS truss · room wash	43.5'	1-323	Extended · 21 ch	A3
204	GLP Impression X4	DS truss · room wash	51.0'	1-344	Extended · 21 ch	A4
205	GLP Impression X4	DS truss · room wash	77.9'	2-281	Extended · 21 ch	B2
206	GLP Impression X4	DS truss · room wash	85.6'	2-302	Extended · 21 ch	B3
207	GLP Impression X4	DS truss · room wash	92.0'	2-323	Extended · 21 ch	B4
208	GLP Impression X4	DS truss · room wash	98.6'	2-344	Extended · 21 ch	B5
209	GLP Impression X4	LED wall · stage wash		3-001	Extended · 21 ch	C1
210	GLP Impression X4	LED wall · stage wash		3-022	Extended · 21 ch	C1
211	GLP Impression X4	LED wall · stage wash		3-043	Extended · 21 ch	C2
212	GLP Impression X4	LED wall · stage wash		3-064	Extended · 21 ch	C2
213	GLP Impression X4	LED wall · stage wash		3-085	Extended · 21 ch	C3
214	GLP Impression X4	LED wall · stage wash		3-106	Extended · 21 ch	C3
215	GLP Impression X4	LED wall · stage wash		3-127	Extended · 21 ch	C4
216	GLP Impression X4	LED wall · stage wash		3-148	Extended · 21 ch	C4
217	GLP Impression X4	LED wall · stage wash		3-169	Extended · 21 ch	C5
218	GLP Impression X4	LED wall · stage wash		3-190	Extended · 21 ch	C5
301	Chauvet COLORado Batten 72X	Floor · house left drape		not patched	TBD	D1
302	Chauvet COLORado Batten 72X	Floor · house left drape		not patched	TBD	D1

UNIT	FIXTURE	POSITION	ON TRUSS	DMX (U-ADDR)	MODE	CIRCUIT
303	Chauvet COLORado Batten 72X	Floor · house left drape		not patched	TBD	**D1**
304	Chauvet COLORado Batten 72X	Floor · house left drape		not patched	TBD	**D1**
305	Chauvet COLORado Batten 72X	Floor · house left drape		not patched	TBD	**D2**
306	Chauvet COLORado Batten 72X	Floor · house left drape		not patched	TBD	**D2**
307	Chauvet COLORado Batten 72X	Floor · house left drape		not patched	TBD	**D2**
308	Chauvet COLORado Batten 72X	Floor · house left drape		not patched	TBD	**D2**
309	Chauvet COLORado Batten 72X	Floor · house right drape		not patched	TBD	**D3**
310	Chauvet COLORado Batten 72X	Floor · house right drape		not patched	TBD	**D3**
311	Chauvet COLORado Batten 72X	Floor · house right drape		not patched	TBD	**D3**
312	Chauvet COLORado Batten 72X	Floor · house right drape		not patched	TBD	**D3**
313	Chauvet COLORado Batten 72X	Floor · house right drape		not patched	TBD	**D6**
314	Chauvet COLORado Batten 72X	Floor · house right drape		not patched	TBD	**D6**
315	Chauvet COLORado Batten 72X	Floor · house right drape		not patched	TBD	**D6**
316	Chauvet COLORado Batten 72X	Floor · house right drape		not patched	TBD	**D6**

Patch from Jeff D'Amato's MagicQ file (Oct 9): Diablo S in Diablo Ex mode (56 ch), Impression X4 in Extended mode (21 ch). Universe 1 = truss house-left half, Universe 2 = truss house-right half, Universe 3 = LED-wall X4s. The 16 COLORado Batten 72X floor units are not patched yet. Unit numbers 301–316 are placeholders.

LABELS

Label both ends of every Soca, every breakout tail and the D3/D6 cross-stage lines. Fixture labels carry unit, DMX address and circuit.

SOCA A
DS truss, house-left half (U1)
Distro end

SOCA A
DS truss, house-left half (U1)
Breakout end

A1
Units 201, 101
900 W · 7.5 A · Ph A

A2
Units 202, 102
900 W · 7.5 A · Ph B

A3
Units 203, 103
900 W · 7.5 A · Ph C

A4
Units 204, 104
900 W · 7.5 A · Ph A

A6
Units 105
550 W · 4.6 A · Ph C

SOCA B
DS truss, house-right half (U2)
Distro end

SOCA B
DS truss, house-right half (U2)
Breakout end

B2
Units 107, 205
900 W · 7.5 A · Ph B

B3
Units 108, 206
900 W · 7.5 A · Ph C

B4
Units 109, 207
900 W · 7.5 A · Ph A

B5
Units 110, 208
900 W · 7.5 A · Ph B

B6
Units 106
550 W · 4.6 A · Ph C

SOCA C
LED wall X4s (U3)
Distro end

SOCA C
LED wall X4s (U3)
Breakout end

C1
Units 209, 210
700 W · 5.8 A · Ph A

C2
Units 211, 212
700 W · 5.8 A · Ph B

C3
Units 213, 214
700 W · 5.8 A · Ph C

C4
Units 215, 216
700 W · 5.8 A · Ph A

C5
Units 217, 218
700 W · 5.8 A · Ph B

SOCA D
All 16 floor COLORados
Distro end

SOCA D
All 16 floor COLORados
Breakout end

D1
Units 301, 302, 303, 304
644 W · 5.4 A · Ph A

D2
Units 305, 306, 307, 308
644 W · 5.4 A · Ph B

D3
Units 309, 310, 311, 312
644 W · 5.4 A · Ph C

D6
Units 313, 314, 315, 316
644 W · 5.4 A · Ph C

D3 LINE
HR drape, units 309–312
Both ends of 100' + 50'

D6 LINE
HR drape, units 313–316
Both ends of 100' + 50'

101
Ayrton Diablo S
DMX 1-001 · A1

102
Ayrton Diablo S
DMX 1-057 · A2

103
Ayrton Diablo S
DMX 1-113 · A3

104
Ayrton Diablo S
DMX 1-169 · A4

105
Ayrton Diablo S
DMX 1-225 · A6

106
Ayrton Diablo S
DMX 2-001 · B6

107
Ayrton Diablo S
DMX 2-057 · B2

108
Ayrton Diablo S
DMX 2-113 · B3

109
Ayrton Diablo S
DMX 2-169 · B4

110
Ayrton Diablo S
DMX 2-225 · B5

201
GLP Impression X4
DMX 1-281 · A1

202
GLP Impression X4
DMX 1-302 · A2

203
GLP Impression X4
DMX 1-323 · A3

204
GLP Impression X4
DMX 1-344 · A4

205
GLP Impression X4
DMX 2-281 · B2

206
GLP Impression X4
DMX 2-302 · B3

207
GLP Impression X4
DMX 2-323 · B4

208
GLP Impression X4
DMX 2-344 · B5

209
GLP Impression X4
DMX 3-001 · C1

210
GLP Impression X4
DMX 3-022 · C1

211
GLP Impression X4
DMX 3-043 · C2

212
GLP Impression X4
DMX 3-064 · C2

213
GLP Impression X4
DMX 3-085 · C3

214
GLP Impression X4
DMX 3-106 · C3

215
GLP Impression X4
DMX 3-127 · C4

216
GLP Impression X4
DMX 3-148 · C4

217
GLP Impression X4
DMX 3-169 · C5

218
GLP Impression X4
DMX 3-190 · C5

301
Chauvet COLORado Batten 72X
DMX not patched · D1

302
Chauvet COLORado Batten 72X
DMX not patched · D1

303
Chauvet COLORado Batten 72X
DMX not patched · D1

304
Chauvet COLORado Batten 72X
DMX not patched · D1

305
Chauvet COLORado Batten 72X
DMX not patched · D2

306
Chauvet COLORado Batten 72X
DMX not patched · D2

307
Chauvet COLORado Batten 72X
DMX not patched · D2

308
Chauvet COLORado Batten 72X
DMX not patched · D2

309
Chauvet COLORado Batten 72X
DMX not patched · D3

310
Chauvet COLORado Batten 72X
DMX not patched · D3

311
Chauvet COLORado Batten 72X
DMX not patched · D3

312
Chauvet COLORado Batten 72X
DMX not patched · D3

313
Chauvet COLORado Batten 72X
DMX not patched · D6

314
Chauvet COLORado Batten 72X
DMX not patched · D6

315
Chauvet COLORado Batten 72X
DMX not patched · D6

316
Chauvet COLORado Batten 72X
DMX not patched · D6